

Mean Field Model

Mean Field Approximation:

K-bit vector: $\vec{x} \in \{0, 1\}^K$

Let $p(t) = P_r[x_j = 1]$ for any $j \in \{1, 2, \dots, K\}$

★ Violation Landscape

→ For AND, both bits should be 1 for No violation

$$p_{\text{AND}}^{(V)} = 1 - p^2$$

Similarly (||ly),

$$p_{\text{XOR}}^{(V)} = (1 - p)^2 + p^2 = 1 - 2p(1 - p)$$

$M = \alpha K$, total clauses, half AND

$$\therefore V(p) = \frac{M}{2}(1 - p^2) + \frac{M}{2}(1 - 2p(1 - p))$$

$$V(p) = M \left(\frac{1 - p + p^2}{2} \right)$$

Stationary point / Attractor:

$$\frac{dV(p)}{dp} = 0 \Rightarrow M(p - 1) = 0$$

$$p^* = 1$$

$$V^* = V(p^*) = V(1) = \frac{M}{2} = \boxed{\frac{\alpha K}{2}}$$

Finding violations after flipping

A typical Clause can be written as : $C_k = (x_i, x_j, z_k)$; where $z_k \in \{\text{AND}, \text{XOR}\}$ is the type of clause and x_i and x_j are variables.

- total clauses: $M = \alpha K$
- A variable x_i occurs in $\left(\frac{2M}{K}\right)$ clauses on average $\rightarrow 2\alpha$

- $\therefore$ Any variable x_i participates (on average) in α AND clauses and α XOR clauses.

→ **What happens on a flip?**

Say we are considering one variable x_j

Say $x_j(t) = 0$, & we try flipping it to 1

in the algorithm, we check # of violations for j

say, $\Delta = V_{t+1} - V_t$

if $\Delta < 0$, flip is beneficial & we keep it, otherwise we discard it, this is our greedy hill climbing algorithm.

$\Rightarrow$ if $C = (j, k, \text{AND})$, is a clause that x_j is involved in. i.e. an AND clause between j & k . And we are flipping j from $0 \rightarrow 1$

→ The probability of x_k being 1 = p

$\therefore \Delta = -1$ with probability p

and with probability $1 - p$, $x_k = 0$, $x_j(0 \rightarrow 1)$ stays violated in this case

Therefore the expectation value of the violation change in this flip for an AND clause is

$$\Rightarrow E_{\text{AND}}[\Delta(x_j : 0 \rightarrow 1)] = p(-1) + 0(1 - p) = -p$$

Similarly (||ly) For XOR $x_j : 0 \rightarrow 1$

if $x_k = 0$, $(0, 0) \rightarrow (0, 1)$ $\Delta = +1$

if $x_k = 1$, $(0, 1) \rightarrow (1, 1)$ $\Delta = -1$

$$\therefore E_{\text{XOR}}[\Delta(x_j : 0 \rightarrow 1)] = 1(p) - 1(1 - p) = 2p - 1$$

Now, we sum over all clauses touching that one variable x_j , since on average we noticed that one variable is involved in α AND clauses and α XOR clauses.

$$\Delta_{0 \rightarrow 1} \simeq \alpha(-p) + \alpha(2p - 1) = -\alpha(1 - p)$$

Therefore, **in the mean field approximation, flipping a bit/variable from 0 to 1 is always beneficial as it decreases the violations.** Unless ofcourse when $p = 1$, when all variables are one and there is no meaning to the flip.

Now, if we do the same study for the reverse flip $x_j : 1 \rightarrow 0$:

For the AND clause,

if $x_k = 0$, $(1, 0) \rightarrow (0, 0)$ $\Delta = 0$

if $x_k = 1, (1, 1) \rightarrow (1, 0) \quad \Delta = +1$

Thus, for AND

$$\Rightarrow E_{\text{AND}}[\Delta(x_j : 1 \rightarrow 0)] = p(+1) + 0(1 - p) = p$$

for XOR:

if $x_k = 0, (1, 0) \rightarrow (0, 0) \quad \Delta = 1$

if $x_k = 1, (1, 1) \rightarrow (1, 0) \quad \Delta = -1,$

Thus the expected violation count:

$$\Rightarrow E_{\text{AND}}[\Delta(x_j : 1 \rightarrow 0)] = p(-1) + 1(1 - p) = 1 - 2p$$

So, finally

$$\Delta_{1 \rightarrow 0} \simeq \alpha(p) + \alpha(1 - 2p) = \alpha(1 - p)$$

Thus, by flipping a variable from one to zero, is on average never good, as it increases violation count.

Combining these two facts, we can see how (atleast in the mean field approximation), flipping all variables to 1 is the attractor for the system. This is also what we got when we did the attractor analysis from dV/dt equation above.

Finding rate equation

Let the rate at which an agent learns a new clause be r . In our code, an agent learns a new clause in two ways, by observing the environment and by talking to her neighbour.

Let us say r is the combined rate of learning a new clause. When an agent learns a new clause, it contains two variables out of total K variables.

Fraction of variables with 0 value is $1 - p$.

The rate of change of p is proportional to :

- The number of zeros that are in the beliefs of the agent. (Since flipping of zero to one is the only change that gets approved, violation goes down)
- The number of variables that we obtain from the clause for which we check violations.
- And the rate of learning r .

$$\frac{dp}{dt} \propto \frac{2}{K} r (1 - p) = \frac{2\gamma}{K} r (1 - p)$$

We can solve this differential equation:

$$\begin{aligned}\frac{dp}{1-p} &= cdt \\ -\ln(1-p) &= ct + c_1 \\ 1-p(t) &= (1-p_0)e^{-ct}\end{aligned}$$

$$p(t) = 1 - (1-p_0)e^{-(2\gamma r/K)t}$$

Thus, we can say, $p(t)$ approaches exponentially with a timescale $\tau = \frac{K}{2\gamma r}$

Here, γ is a proportionality constant, r is the rate of adding clauses and K is the number of variables. Larger K slows convergence, while more frequent learning speeds convergence.

Homogeneity in mean field approximation

In our code, we calculate homogeneity using

```
def compute_homogeneity(self):
    if self.N == 0: return 0
    total = sum(max(sum(1 for a in self.agents if a.x[j] == 1),
self.N - sum(1 for a in self.agents if a.x[j] == 1)) / self.N for j
in range(self.K))
    return total / self.K
```

This can be written as

$$H(t) = \frac{1}{K} \sum_{j=1}^K \frac{\max(n_j(t), N - n_j(t))}{N},$$

where $n_j(t)$ is the number of agents whose j -th bit/variable equals 1.

In the mean field approximation, all coordinates are statistically equivalent and N is large, then n_j/N is close to a common fraction $p(t)$, so

$$H(t) \approx \max(p(t), 1 - p(t)).$$

Our dynamics pushes our model to $p^* = 1$, thus once $p(t) > 1/2$, we follow $H(t) \approx p(t)$.

From our mean field approximation:

$$\frac{dp}{dt} = \frac{2r\gamma}{K}(1-p),$$

Since $H = \max(p, 1 - p)$, its derivative is defined with cases:

$$\frac{dH}{dt} = \begin{cases} -\frac{dp}{dt}, & p < \frac{1}{2}, \\ [4pt] \frac{dp}{dt}, & p > \frac{1}{2}. \end{cases}$$

$$\left| \frac{dH}{dt} \right| \simeq \frac{2r\gamma}{K} (1 - H).$$

Thus, homogeneity moves to its attractor which is also 1 at the same rate as p .
With the solution:

$$H(t) \approx 1 - (1 - H_0)e^{-(2r\gamma/K)t}.$$

We should get an inflection point at $p = 1/2$ for our basic model.

This is why our homogeneity curves are way smoother than the violation curves.

- r is the rate of finding new clauses, it should include both observation and interaction. $r = r_{obs} + r_{int}$.
- r_{int} we expect to be proportional to the average in-degree of the network, whereas r_{obs} is an external parameter.

p=1 is not the attractor

We know that $p = 1$ is not the attractor for our dynamics. As it would push homogeneity to 1. In our simulation we notice it converges smoothly to different values for different networks.

- **! We need a better mean field model!**

Pair Mean-Field Extension of the Mean Field Theory

The old mean-field theory keeps only the scalar

$$p(t) = \Pr(x_i = 1),$$

and then factorizes all two-bit probabilities in terms of p

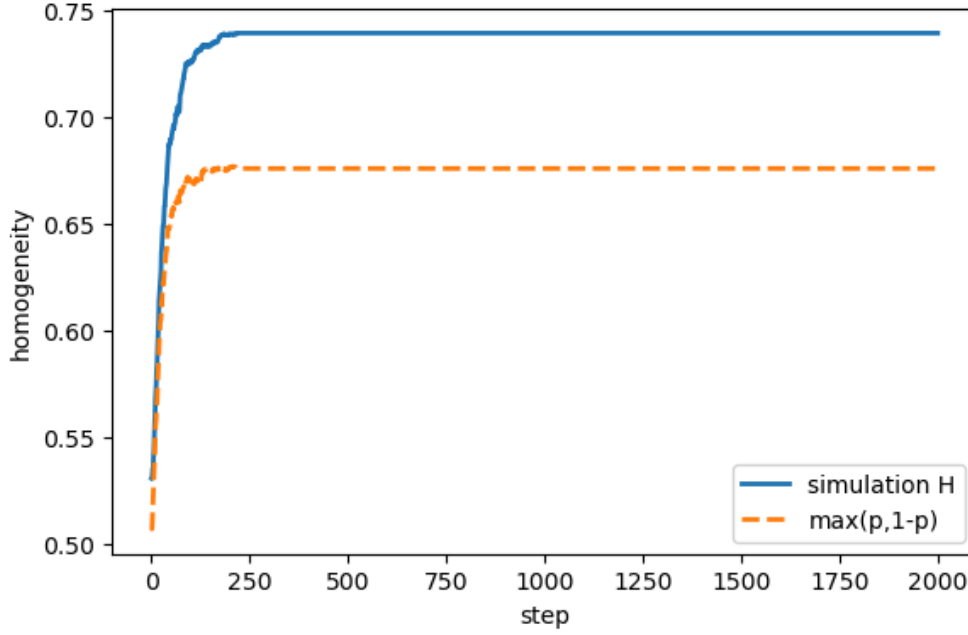
This works only if:

1. all coordinates are statistically equivalent, and

2. there are no important pair correlations between variables that co-occur in clauses.

Numerical analysis shows the following:

- $q_{11}^{\text{AND,actual}} \neq p^2$, where q_{11} is the probability of both variables in a clause being one.
- $q_{\text{eq}}^{\text{XOR,actual}} \neq p^2 + (1 - p)^2$,
- $H \neq \max(p, 1 - p)$ once the coordinate frequencies n_j broaden.



Therefore the new mean-field theory should keep not only the one-point quantity p , but also suitable two-point observables, i.e. for a clause

Exact pair-level observables

Let us now divide the total clauses (M) into both categories, let

$$M_A = \#(\text{AND clauses}), \quad M_X = \#(\text{XOR clauses}), \quad M = M_A + M_X.$$

For a clause $C_k = (i, j, q_k)$, we can now define the pair probabilities across agents:

$$q_{ab}^{(ij)}(t) = \Pr(x_i = a, x_j = b), \quad a, b \in \{0, 1\}.$$

For an AND clause, the clause is satisfied only when both bits are 1. Hence the violation probability is

$$v_A^{(ij)} = 1 - q_{11}^{(ij)}. \quad (1)$$

For an XOR clause in the code, the clause is satisfied when the parity is odd, i.e. when the two variables are different. Hence the violation probability is

$$v_X^{(ij)} = q_{00}^{(ij)} + q_{11}^{(ij)}. \quad (2)$$

Define the equality probability for XOR clauses (probability where both variables are equal and thus violate the clause):

$$q_{\text{eq}}^{(ij)} = q_{00}^{(ij)} + q_{11}^{(ij)}.$$

Then

$$v_X^{(ij)} = q_{\text{eq}}^{(ij)}. \quad (3)$$

Now average over the clause ensemble:

$$q_{11}^A = \frac{1}{M_A} \sum_{(i,j) \in A} q_{11}^{(ij)}, \quad q_{\text{eq}}^X = \frac{1}{M_X} \sum_{(i,j) \in X} q_{\text{eq}}^{(ij)}.$$

Then the exact mean violation count per agent is

$$V(t) = M_A(1 - q_{11}^A(t)) + M_X q_{\text{eq}}^X(t). \quad (4)$$

This is an exact identity, not an approximation, like the previous one.

Exact identity for homogeneity

Let

$$f_j(t) = \frac{1}{N} \sum_{\alpha=1}^N x_j^{(\alpha)}(t)$$

be the fraction of agents with bit $j = 1$.

The model computes homogeneity as scalar

$$H(t) = \frac{1}{K} \sum_{j=1}^K \max(f_j(t), 1 - f_j(t)). \quad (5)$$

We can use $\max(a, b) = \frac{a+b}{2} + \frac{|a-b|}{2}$,

$$H(t) = \frac{1}{2} + \frac{1}{K} \sum_{j=1}^K |f_j(t) - 1/2|. \quad (6)$$

So H is not determined by the global mean as I had derived before

$$p(t) = \frac{1}{K} \sum_{j=1}^K f_j(t)$$

unless all f_j are tightly concentrated around a common value.

Thus the old result

$$H \approx \max(p, 1 - p)$$

is only valid in the additional limit where the distribution of f_j is narrow.

Exact local violation change for a proposed flip

Now consider a proposed flip of one variable x_j in one agent.

Write

$$\Delta V_j = V_j^{\text{new}} - V_j^{\text{old}},$$

where V_j is the violation count on the relevant local set of clauses that the algorithm checks during the update.

From the code, for a newly learned clause containing variable j , the model constructs:

- `related_kb` : all clauses in the current KB involving either variable from the newly learned clause,
- `affected_kb(j)` : all clauses in `related_kb` that contain j ,

and accepts the flip iff

$$\Delta V_j < 0. \tag{7}$$

So the exact acceptance event is

$$\mathbf{1}_{\text{accept}}(j) = \mathbf{1}\{\Delta V_j < 0\}. \tag{8}$$

Now decompose ΔV_j by clause type around j .

Let the local counts around j be:

- $A_1(j)$: number of AND clauses touching j whose partner bit is 1,
- $A_0(j)$: number of AND clauses touching j whose partner bit is 0,
- $X_1(j)$: number of XOR clauses touching j whose partner bit is 1,
- $X_0(j)$: number of XOR clauses touching j whose partner bit is 0.

Then the change in local violations under a flip can be written exactly.

Flip $0 \rightarrow 1$

Suppose $x_j = 0$, and we test $x_j : 0 \rightarrow 1$.

AND contribution

For an AND clause touching j :

- if partner = 1, then the clause changes from violated to satisfied, so contribution = -1 ,
- if partner = 0, then the clause stays violated, so contribution = 0.

Hence

$$\Delta V_j^A(0 \rightarrow 1) = -A_1(j). \quad (9)$$

XOR contribution

For an XOR clause touching j :

- if partner = 1, then $01 \rightarrow 11$, so satisfied $\rightarrow$ violated, contribution = $+1$,
- if partner = 0, then $00 \rightarrow 10$, so violated $\rightarrow$ satisfied, contribution = -1 .

Hence

$$\Delta V_j^X(0 \rightarrow 1) = X_1(j) - X_0(j). \quad (10)$$

Therefore the total local violation change is

$$\Delta V_j(0 \rightarrow 1) = -A_1(j) + X_1(j) - X_0(j). \quad (11)$$

The flip is accepted iff
our

$$-A_1(j) + X_1(j) - X_0(j) < 0. \quad (12)$$

Equivalently,

$$A_1(j) + X_0(j) > X_1(j). \quad (13)$$

Flip $1 \rightarrow 0$

Suppose $x_j = 1$, and we test $x_j : 1 \rightarrow 0$.

AND contribution

For an AND clause touching j :

- if partner = 1, then the clause changes from satisfied to violated, contribution = +1,
- if partner = 0, then the clause stays violated, contribution = 0.

Hence

$$\Delta V_j^A(1 \rightarrow 0) = +A_1(j). \quad (14)$$

XOR contribution

For an XOR clause touching j :

- if partner = 1, then 11 $\rightarrow$ 01, so violated $\rightarrow$ satisfied, contribution = -1,
- if partner = 0, then 10 $\rightarrow$ 00, so satisfied $\rightarrow$ violated, contribution = +1.

Hence

$$\Delta V_j^X(1 \rightarrow 0) = X_0(j) - X_1(j). \quad (15)$$

Therefore the total local violation change is

$$\Delta V_j(1 \rightarrow 0) = A_1(j) + X_0(j) - X_1(j). \quad (16)$$

The flip is accepted iff

$$A_1(j) + X_0(j) - X_1(j) < 0. \quad (17)$$

Equivalently,

$$X_1(j) > A_1(j) + X_0(j). \quad (18)$$

Numerical verification:

In the following code, I use one simulation run and check all the flips that were accepted. And for those flips I check if they followed the inequality.

```
flip_df["ineq_predict_accept"] = False

mask01 = flip_df["flip_type"] == "0->1"
mask10 = flip_df["flip_type"] == "1->0"

flip_df.loc[mask01, "ineq_predict_accept"] = (
    flip_df.loc[mask01, "A1"] + flip_df.loc[mask01, "X0"] >
    flip_df.loc[mask01, "X1"]
)
```

```

flip_df.loc[mask10, "ineq_predict_accept"] = (
    flip_df.loc[mask10, "X1"] > flip_df.loc[mask10, "A1"] +
    flip_df.loc[mask10, "X0"]
)

(
    flip_df[["flip_type", "accepted", "ineq_predict_accept"]]
        .assign(match=lambda d: d["accepted"] ==
d["ineq_predict_accept"])
        .groupby("flip_type")["match"]
        .mean()
)

```

Output:

```

flip_type
0->1 1.0
1->0 1.0
Name: match, dtype: float64

```

The result 1.0 for both flips type show our inequalities are correct and also show that our first model was not a correct understanding of the model, as that one only had flips from 0 to 1 accepting.

Exact acceptance probabilities

We can now define acceptance conditional probabilities

$$A_{0 \rightarrow 1}(t) = \Pr(\Delta V_j(0 \rightarrow 1) < 0 \mid x_j = 0), \quad (19)$$

$$A_{1 \rightarrow 0}(t) = \Pr(\Delta V_j(1 \rightarrow 0) < 0 \mid x_j = 1). \quad (20)$$

Using the exact formulas obtained above,

$$A_{0 \rightarrow 1}(t) = \Pr(A_1 + X_0 > X_1 \mid x_j = 0), \quad (21)$$

$$A_{1 \rightarrow 0}(t) = \Pr(X_1 > A_1 + X_0 \mid x_j = 1). \quad (22)$$

These are exact but still not closed, because they depend on the full joint distribution of the local counts A_1, X_0, X_1 . **And importantly these depend on the statistics of clauses**

Dynamics of the mean violation count

Once the pair observables are known,

$$V(t) = M_A(1 - q_{11}^A(t)) + M_X q_{eq}^X(t). \quad (23)$$

Therefore

$$\frac{dV}{dt} = -M_A \frac{dq_{11}^A}{dt} + M_X \frac{dq_{eq}^X}{dt}. \quad (24)$$

So the problem reduces to finding closed equations for

$$\frac{dq_{11}^A}{dt}, \quad \frac{dq_{eq}^X}{dt}. \quad (25)$$

At pair level these equations depend on triple statistics, because a flip at j changes all pairs (i, j) attached to j . It was easy to write dp/dt , but this would depend on the flips and other conditional statistics.

For now, equations (23) and (24) already show the key structural difference from the old mean field:

- in the old theory, V is a function of p alone,
- in the new theory, V depends on independent pair observables.

Lowest-order correction around the old scalar mean field

The old theory is recovered by factorization:

$$q_{11}^A \approx p^2, \quad q_{eq}^X \approx p^2 + (1 - p)^2. \quad (26)$$

Then

$$V \approx M_A(1 - p^2) + M_X(p^2 + (1 - p)^2). \quad (27)$$

If $M_A = M_X = M/2$, this becomes

$$V \approx M \left(1 - p + \frac{p^2}{2} \right), \quad (28)$$

which is exactly the old simpler mean field expression.

So the new theory is not a different model; it is a strict extension of the old mean-field theory where factorization is no longer imposed at the pair level.